Package 'betaLB'

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Title Likelihood-Based Confidence Intervals for Regression Effect Sizes

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Description Generates profile likelihood confidence intervals for standardized regression coefficients (beta) and other effect sizes, including multiple correlation, semipartial correlations, improvement in R-squared, squared partial correlations, differences in standardized regression coefficients, and Cohen's f-squared, for models fitted by lm().

URL https://github.com/jeksterslab/betaLB, https://jeksterslab.github.io/betaLB/

BugReports https://github.com/jeksterslab/betaLB/issues

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`BetaLB` *Likelihood Intervals for Standardized Regression Slopes*

Description

Extract already-computed point estimates and intervals from an LB context. For multi-coefficient nonnegative effects, vector projection is conservative.

Usage

```
BetaLB(object)
```

Arguments

`object`	Context returned by `LB()`, with this family computed.

Value

Object of class `betalb`; use `coef()`, `confint()` or `summary()`.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Likelihood Effect Extractors: `DeltaRSqLB()`, `DiffBetaLB()`, `FSqLB()`, `PCorLB()`, `RSqLB()`, `SCorLB()`

Examples

```
fit <- stats::lm(formula = mpg ~ wt + hp, data = mtcars)
out <- LB(object = fit, effects = "beta", type = "mvn", alpha = 0.05)
BetaLB(object = out)
```

`coef.betalb` *Estimated Parameter Method for an Object of Class* `betalb`

Description

Estimated Parameter Method for an Object of Class `betalb`

Usage

```
## S3 method for class 'betalb'
coef(object, ...)
```

Arguments

`object`	Object of class `betalb`.
`...`	Additional arguments.

Value

Returns a vector of estimated parameters.

Author(s)

Ivan Jacob Agaloos Pesigan

Examples

```
object <- structure(
  list(est = c(x = 0.5), args = list(type = "mvn", alpha = 0.05)),
  class = "betalb"
)
stats::coef(object = object)
```

`confint.betalb` *Confidence Intervals Method for an Object of Class* `betalb`

Description

Confidence Intervals Method for an Object of Class `betalb`

Usage

```
## S3 method for class 'betalb'
confint(object, parm = NULL, level = 0.95, type = NULL, ...)
```

Arguments

`object`	Object of class `betalb`.
`parm`	A specification of which parameters are to be given confidence intervals.
`level`	Confidence level of stored intervals. Only already-computed levels are available.
`type`	Stored calibration type; `NULL` selects the first type.
`...`	Additional arguments.

Value

Returns a matrix of confidence intervals.

Author(s)

Ivan Jacob Agaloos Pesigan

Examples

```
object <- structure(
  list(est = c(x = 0.5), args = list(type = "mvn", alpha = 0.05)),
  class = "betalb"
)
try(stats::confint(object = object))
```

DeltaRSqLB *Likelihood Intervals for Increments in R-Squared*

Description

Extract already-computed point estimates and intervals from an LB context. For multi-coefficient nonnegative effects, vector projection is conservative.

Usage

```
DeltaRSqLB(object)
```

Arguments

object	Context returned by LB(), with this family computed.

Value

Object of class betalb; use coef(), confint() or summary().

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Likelihood Effect Extractors: BetaLB(), DiffBetaLB(), FSqLB(), PCorLB(), RSqLB(), SCorLB()

Examples

```
fit <- stats::lm(formula = mpg ~ wt + hp, data = mtcars)
out <- LB(object = fit, effects = "deltarsq", type = "mvn", alpha = 0.05)
DeltaRSqLB(object = out)
```

DiffBetaLB *Likelihood Intervals for Differences of Standardized Slopes*

Description

Extract already-computed point estimates and intervals from an LB context. For multi-coefficient nonnegative effects, vector projection is conservative.

Usage

```
DiffBetaLB(object)
```

Arguments

object	Context returned by LB(), with this family computed.

Value

Object of class betalb; use coef(), confint() or summary().

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Likelihood Effect Extractors: BetaLB(), DeltaRSqLB(), FSqLB(), PCorLB(), RSqLB(), SCorLB()

Examples

```
fit <- stats::lm(formula = mpg ~ wt + hp, data = mtcars)
out <- LB(object = fit, effects = "diffbeta", type = "mvn", alpha = 0.05)
DiffBetaLB(object = out)
```

FSqLB *Likelihood Intervals for Overall and Local F-Squared*

Description

Extract already-computed point estimates and intervals from an LB context. For multi-coefficient nonnegative effects, vector projection is conservative.

Usage

```
FSqLB(object)
```

Arguments

object	Context returned by LB(), with this family computed.

Value

Object of class betalb; use coef(), confint() or summary().

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Likelihood Effect Extractors: BetaLB(), DeltaRSqLB(), DiffBetaLB(), PCorLB(), RSqLB(), SCorLB()

Examples

```
fit <- stats::lm(formula = mpg ~ wt + hp, data = mtcars)
out <- LB(object = fit, effects = "fsq", type = "mvn", alpha = 0.05)
FSqLB(object = out)
```

LB *Likelihood Intervals for Regression Effect Sizes*

Description

Fit covariance-likelihood regions for independently sampled random regressors. Compute requested effects, types and confidence levels in one native batch, then use the effect extractors to select stored results.

Usage

```
LB(
  object,
  type = c("mvn", "robust"),
  alpha = c(0.05, 0.01, 0.001),
  effects = c("beta", "rsq", "scor", "pcor", "deltarsq", "diffbeta", "fsq"),
  std_y = TRUE,
  std_x = TRUE,
  blocks = list(),
  control = list()
)
```

Arguments

object	Unweighted, full-rank scalar-response lm fit with an intercept, stored model frame, no offsets and positive residual variance.
type	Character vector containing mvn, robust, or both.
alpha	Distinct significance levels in (0, 1).
effects	Requested families: beta, rsq, scor, pcor, deltarsq, diffbeta, and fsq.
std_y	Whether to standardize the response for slope targets.
std_x	Logical flags or names of model-matrix predictors to standardize for slope targets. Other effects keep their definitions.
blocks	Named list of additional predictor blocks, identified by model-matrix column names or indices, for increments and local f-squared.
control	Named list: maxeval is the native evaluation budget per endpoint; tolerance is its normalized verification tolerance.

Details

Normal theory uses divisor-n Gaussian covariance likelihood. Robust calibration uses empirical fourth moments. Signed scalar targets use a target-specific variance ratio; multi-coefficient nonnegative effects use conservative vector projection and a largest-eigenvalue robust cutoff. These are asymptotic procedures, without a finite-sample coverage guarantee.

Value

Context of class `lb`, containing named point estimates, a long `intervals` table and per-row endpoint `diagnostics`. Failed endpoints are `NA` with a specific status. Extractors return `betalb` objects. Slope/contrast target IDs retain their standardization flags.

Author(s)

Ivan Jacob Agaloos Pesigan

Examples

```
fit <- stats::lm(formula = mpg ~ wt, data = mtcars)
out <- LB(object = fit, type = "mvn", alpha = 0.05, effects = "rsq")
RSqLB(object = out)
```

---

nas1982 *1982 National Academy of Sciences Doctoral Programs Data*

Description

1982 National Academy of Sciences Doctoral Programs Data

Usage

```
nas1982
```

Format

Ratings of 46 doctoral programs in psychology in the USA with the following variables:

QUALITY Program quality ratings.

NFACUL Number of faculty members in the program.

NGRADS Number of program graduates.

PCTSUPP Percentage of program graduates who received support.

PCTGRT Percent of faculty members holding research grants.

NARTIC Number of published articles attributed to program faculty member.

PCTPUB Percent of faculty with one or more published article.

References

National Research Council. (1982). *An assessment of research-doctorate programs in the United States: Social and behavioral sciences*. doi:10.17226/9781. Reproduced with permission from the National Academy of Sciences, Courtesy of the National Academies Press, Washington, D.C.

`PCorLB` *Likelihood Intervals for Squared Partial Correlations*

Description

Extract already-computed point estimates and intervals from an LB context. For multi-coefficient nonnegative effects, vector projection is conservative.

Usage

```
PCorLB(object)
```

Arguments

`object`	Context returned by `LB()`, with this family computed.

Value

Object of class `betalb`; use `coef()`, `confint()` or `summary()`.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Likelihood Effect Extractors: `BetaLB()`, `DeltaRSqLB()`, `DiffBetaLB()`, `FSqLB()`, `RSqLB()`, `SCorLB()`

Examples

```
fit <- stats::lm(formula = mpg ~ wt + hp, data = mtcars)
out <- LB(object = fit, effects = "pcor", type = "mvn", alpha = 0.05)
PCorLB(object = out)
```

`print.betalb` *Print Method for an Object of Class* `betalb`

Description

Print Method for an Object of Class `betalb`

Usage

```
## S3 method for class 'betalb'
print(x, alpha = NULL, digits = 4, type = NULL, ...)
```

Arguments

`x`	Object of class `betalb`.
`alpha`	Numeric vector. Significance level α. If `alpha = NULL`, show all stored significance levels.
`digits`	Digits to print.
`type`	Stored calibration type; `NULL` selects all for printing.
`...`	Additional arguments.

Value

Prints a matrix of estimates and confidence intervals.

Author(s)

Ivan Jacob Agaloos Pesigan

Examples

```
object <- structure(
  list(est = c(x = 0.5), args = list(type = "mvn", alpha = 0.05)),
  class = "betalb"
)
print(x = object)
```

print.lb *Methods for a Batched Likelihood Context*

Description

Display or select already-computed results without running the solver.

Usage

```
## S3 method for class 'lb'
print(x, alpha = NULL, digits = 4, type = NULL, ...)

## S3 method for class 'lb'
summary(object, alpha = NULL, digits = 4, type = NULL, ...)

## S3 method for class 'lb'
coef(object, ...)

## S3 method for class 'lb'
confint(object, parm = NULL, level = 0.95, type = NULL, ...)

## S3 method for class 'lb'
vcov(object, ...)
```

Arguments

x	Object of class lb.
alpha	Stored significance levels; NULL selects all.
digits	Digits displayed by print methods.
type	Stored calibration type; NULL selects the first type.
...	Arguments forwarded to the corresponding effect method.
object	Object of class lb.
parm	Parameter names or positive row indices; NULL selects all.
level	Confidence level already computed by LB().

Value

Print methods invisibly return their input; summary returns stored interval rows; coef returns point estimates; confint returns a two-column matrix. vcov stops because likelihood intervals do not define a covariance.

Author(s)

Ivan Jacob Agaloos Pesigan

Examples

```
fit <- stats::lm(formula = mpg ~ wt, data = mtcars)
out <- LB(object = fit, effects = "rsq", type = "mvn", alpha = 0.05)
stats::coef(object = out)
stats::confint(object = out)
```

RSqLB *Likelihood Intervals for R-Squared and Its Finite-Sample Adjustment*

Description

Extract already-computed point estimates and intervals from an LB context. For multi-coefficient nonnegative effects, vector projection is conservative.

Usage

```
RSqLB(object)
```

Arguments

`object`	Context returned by `LB()`, with this family computed.

Value

Object of class `betalb`; use `coef()`, `confint()` or `summary()`.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Likelihood Effect Extractors: `BetaLB()`, `DeltaRSqLB()`, `DiffBetaLB()`, `FSqLB()`, `PCorLB()`, `SCorLB()`

Examples

```
fit <- stats::lm(formula = mpg ~ wt + hp, data = mtcars)
out <- LB(object = fit, effects = "rsq", type = "mvn", alpha = 0.05)
RSqLB(object = out)
```

SCorLB *Likelihood Intervals for Signed Semipartial Correlations*

Description

Extract already-computed point estimates and intervals from an LB context. For multi-coefficient nonnegative effects, vector projection is conservative.

Usage

```
SCorLB(object)
```

Arguments

object	Context returned by LB(), with this family computed.

Value

Object of class betalb; use coef(), confint() or summary().

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Likelihood Effect Extractors: BetaLB(), DeltaRSqLB(), DiffBetaLB(), FSqLB(), PCorLB(), RSqLB()

Examples

```
fit <- stats::lm(formula = mpg ~ wt + hp, data = mtcars)
out <- LB(object = fit, effects = "scor", type = "mvn", alpha = 0.05)
SCorLB(object = out)
```

SCorSqLB *Squared Signed-Semipartial Confidence Intervals*

Description

Square the signed interval as a set, with lower limit zero if it contains zero. This differs from an interval of squared draws. No draws are generated and no endpoints are clipped to one. Failed signed endpoints remain missing.

Usage

```
SCorSqLB(object, level = 0.95, type = NULL)
```

Arguments

object	Context accepted by SCorLB().
level	Single confidence level in (0, 1).
type	Stored likelihood calibration, or NULL for the first.

Value

Data frame with est, lower and upper columns and predictor row names. The point estimate is the squared signed estimate. No standard error is implied by this interval transformation.

Author(s)

Ivan Jacob Agaloos Pesigan

Examples

```
fit <- stats::lm(formula = mpg ~ wt + hp, data = mtcars)
out <- LB(object = fit, effects = "scor", type = "mvn", alpha = 0.05)
SCorSqLB(object = out)
```

---

summary.betalb *Summary Method for an Object of Class* betalb

---

Description

Summary Method for an Object of Class betalb

Usage

```
## S3 method for class 'betalb'
summary(object, alpha = NULL, digits = 4, type = NULL, ...)
```

Arguments

object	Object of class betalb.
alpha	Numeric vector. Significance level α. If alpha = NULL, show all stored significance levels.
digits	Digits to print.
type	Stored calibration type; NULL selects all.
...	Additional arguments.

Value

Returns a matrix or list of estimates and confidence intervals.

Author(s)

Ivan Jacob Agaloos Pesigan

Examples

```
object <- structure(
  list(est = c(x = 0.5), args = list(type = "mvn", alpha = 0.05)),
  class = "betalb"
)
summary(object = object)
```

`vcov.betalb` *Sampling Variance-Covariance Matrix Method for an Object of Class* `betalb`

Description

Sampling Variance-Covariance Matrix Method for an Object of Class `betalb`

Usage

```
## S3 method for class 'betalb'
vcov(object, ...)
```

Arguments

`object`	Object of class `betalb`.
`...`	Additional arguments.

Value

Stops with an error because profile likelihood intervals do not define a covariance matrix.

Author(s)

Ivan Jacob Agaloos Pesigan

Examples

```
object <- structure(
  list(est = c(x = 0.5), args = list(type = "mvn", alpha = 0.05)),
  class = "betalb"
)
try(stats::vcov(object = object))
```

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